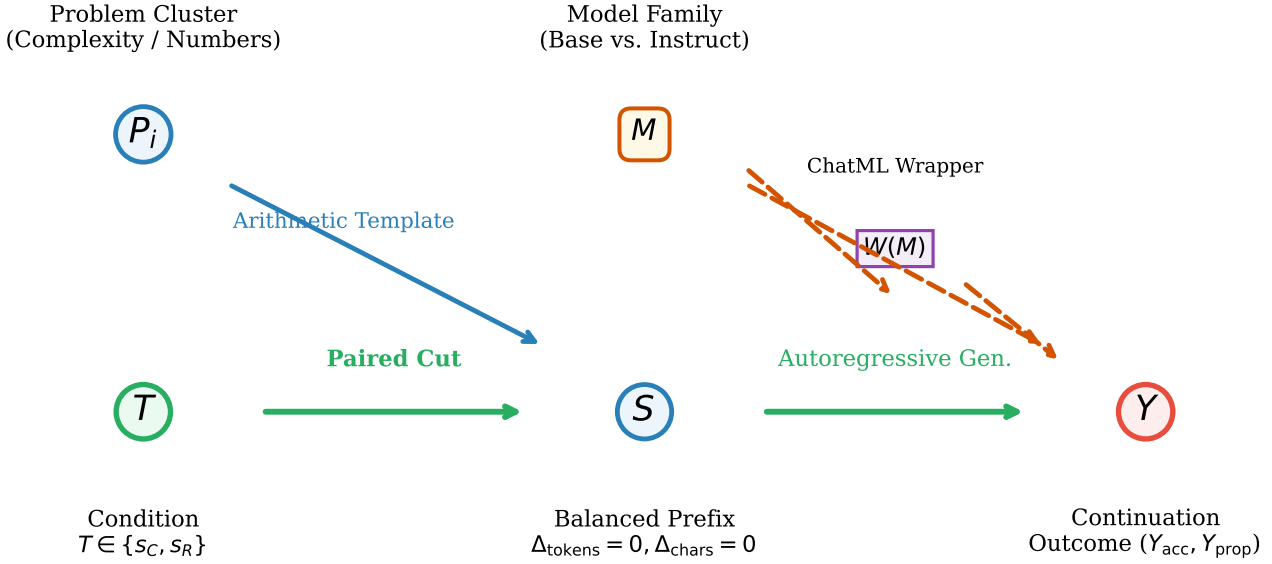


Causal Identifiability Status:

• Within-Model Treatment Effect ($T \to Y \mid M$): CAUSALLY IDENTIFIED Average Treatment Effect (ATE).
Condition $T$ assigned in fully crossed pairs across all $P_i$; $\Delta_{\text{tokens}} = 0, \Delta_{\text{chars}} = 0$ eliminates length confounding.

• Cross-Model Effect ($M \to Y$): OBSERVATIONAL COMPARISON. Checkpoint differs in training and interface ($W(M)$).